Targeted Sensor Placement for Salinity Fields

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Abstract

Monitoring salinity along estuaries requires careful sensor placement, for which greedy acquisition-function designs are the standard approach. However, their optimality guarantees apply to local information-theoretic objectives rather than to the downstream reconstruction error within the scientifically critical sub-region. We introduce the *Dataset Quality Score* (DQS), a scale-free $[0, 1]$ metric that benchmarks any sensor design against an arc-length-equidistant geometric reference on a gradient-weighted RMSE in the steep region.

We consider a logarithmic test function calibrated to the Sooke estuary and perform reconstruction using a Gaussian process with a Matérn-3/2 kernel. Four acquisition functions and three illustrative baselines are evaluated against a global differential-evolution optimiser on sensor counts $N \in \{3, \ldots, 22\}$ with observation noise $\sigma = 0.9$ PSU. Results are averaged over 30 seeds for the greedy methods and 10 for the optimiser, and robustness is verified on two alternative phantom profiles in addition to the calibrated logarithmic fit.

The noise-averaged optimiser attains DQS $= 0.937$ at $N = 20$, while the strongest greedy heuristic (MaxDist) reaches only 0.675—a structural gap of 0.26 DQS, or roughly 28% of achievable quality. The gap peaks at 0.47 in the mid-budget regime ($N = 10$) and narrows as MaxDist gradually fills the halocline. The noiseless optimiser saturates at DQS $= 1$ from $N = 6$, bounding how few sensors suffice when placed optimally.

The framework generalises to any field whose scientific value concentrates in an identifiable sub-region and quantifies the cost of greedy one-step-ahead selection against a jointly optimised reference.

1. Introduction

1.1 Motivation

As a vital resource, water underpins the survival of organisms and the integrity of habitats in any given system. However, in Canada, the frequency of events that compromise the availability and quality of water is increasing. Climate change, through rising sea levels, altered nutrient distribution, and increasing water temperatures, presents unprecedented challenges for water management. Anthropogenic pollution promotes harmful algal blooms and disrupts aquatic ecosystems through chemical contamination Asresu et al. (2025); O'Neil et al. (2013); forest fires introduce contaminants into drainage systems, while floods wash sediments, nutrients, and pollutants into bodies of water Bladon et al. (2014). These threats highlight the growing need to monitor and characterise water quality, both to safeguard ecosystems and to respond to environmental change. Continuous monitoring of salinity, in particular, is essential for the early detection of saltwater intrusion and shifts in estuarine mixing, but is constrained by sensor cost, maintenance, and data-link bandwidth. Operators typically deploy between three and twenty fixed-point probes along a channel spanning several kilometres. The practical question is therefore: *where should the next sensor go?*

1.2 Problem statement

Given a one-dimensional domain $\mathcal{X} \subset \mathbb{R}$ and a target sensor count N, the task is to choose locations $S = \{x_1, \ldots, x_N\} \subset \mathcal{X}$ that maximise the quality of a field $\hat{f}_S$ reconstructed from noisy measurements $y_i = f(x_i) + \varepsilon_i$. Because reconstruction quality is application-specific, the choice of scoring metric is itself a design decision: in estuarine monitoring the scientifically relevant quantity is the accuracy of $\hat{f}_S$ inside the steep region, while errors on the flat offshore plateau carry less weight.

1.3 Contributions

This paper makes five contributions.

1. We define the *Dataset Quality Score* (DQS), a $[0, 1]$-valued metric that normalises a problem-specific gradient-weighted RMSE against an arc-length-equidistant geometric benchmark.
2. We position DQS within the targeted-IMSE tradition of Williams et al. (2000), Picheny et al. (2010), and Bect et al. (2012), and within the regret-vs-best-known convention of Frazier (2018).
3. We empirically quantify the gap between greedy and globally optimised designs on a 1-D salinity test function calibrated to the Sooke estuary, separating noiseless ceilings from noise-averaged realities.
4. We verify the robustness of the resulting rankings across three phantom profiles—a calibrated logarithmic fit, a hyperbolic-tangent halocline, and a piecewise-linear ramp—

to confirm that the conclusions are properties of the criterion–metric alignment rather than of any single analytic choice.

5. We report calibration diagnostics (NLPD, 1σ coverage) as orthogonal secondary scores, demonstrating that the DQS rankings are not an artefact of miscalibrated posteriors.

1.4 Roadmap

Section 2 reviews the relevant literature on experimental design, Gaussian processes, and acquisition functions. Section 3 formalises the test function, domain, and notation. Section 4 introduces the DQS framework. Section 5 describes the evaluation protocols and computational cost. Section 6 reports the empirical gap between greedy and optimised designs. Section 7 reports calibration diagnostics. Sections 8 and 9 discuss the implications and conclude.

2. Background and Related Work

2.1 The sensor-placement problem

Coastal and estuarine salinity fields exhibit steep gradients near freshwater inflows that relax into weakly varying offshore plateaus (Geyer and MacCready, 2014). Because monitoring budgets cap the number of deployable sensors well below the spatial resolution of the underlying physics, operators face a discrete design problem: choose N locations $S \subset \mathcal{X}$ to maximise the quality of the reconstructed field. The problem is combinatorial ($\binom{|\mathcal{X}|}{N}$ candidate configurations) and, in its exact form, NP-hard (**?**).

2.2 Bayesian experimental design and information theory

Classical optimal experimental design (Fedorov, 1972; Chaloner and Verdinelli, 1995) frames sensor placement as maximisation of an information criterion. The A-, D-, and E-optimal criteria target the expected posterior covariance directly; the more flexible *mutual information* (MI) criterion (Caselton and Zidek, 1984; Krause et al., 2008) targets uncertainty reduction between measured and unmeasured locations,

$$I(S;\bar{S}) \;=\; H(\bar{S}) - H(\bar{S} \mid S), \tag{1}$$

with $\bar{S} = \mathcal{X} \setminus S$ and H differential entropy.

Krause et al. (2008) proved that MI is submodular and monotone under a mild regularity condition, so the greedy algorithm achieves a $(1 - 1/e)$ approximation of the optimal MI:

$$I(S_{\text{greedy}}) \;\geq\; \left(1 - \tfrac{1}{e}\right) I(S^*). \tag{2}$$

Equation (2) underpins most acquisition-function methods. Crucially, however, the guarantee concerns the information-theoretic objective rather than a downstream reconstruction metric, and it leaves open a gap of up to $\approx 37\%$ that a global search might close.

2.3 Gaussian-process surrogates

The Gaussian process (GP) (Rasmussen and Williams, 2006) plays two roles here: (i) the reconstruction operator used to score any sensor set, and (ii) the generative model whose posterior variance defines the classical acquisition functions. A GP prior $f \sim \mathcal{GP}(\mu, k)$ with observations $\mathbf{y} = \mathbf{f} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I})$, yields the posterior

$$\mu_S(x) = \mu(x) + \mathbf{k}_{xS}\big[\mathbf{K}_{SS} + \sigma_n^2 \mathbf{I}\big]^{-1}(\mathbf{y} - \boldsymbol{\mu}_S), \tag{3}$$

$$\sigma_S^2(x) = k(x,x) - \mathbf{k}_{xS}\big[\mathbf{K}_{SS} + \sigma_n^2 \mathbf{I}\big]^{-1}\mathbf{k}_{Sx}. \tag{4}$$

The Matérn-3/2 kernel

$$k(x,x') = \sigma_f^2\Big(1 + \tfrac{\sqrt{3}r}{\ell}\Big)\exp\Big(-\tfrac{\sqrt{3}r}{\ell}\Big), \qquad r = \|x - x'\|, \tag{5}$$

is widely used for environmental data because it is once mean-square differentiable, matching the mild roughness of geophysical fields without imposing the smoothness of the squared-exponential kernel (Stein, 1999).

Figure 1 illustrates the effective correlation $K(x_{\mathrm{ref}}, x)$ used in this work. Modelling salinity in $\ln x$ rather than x (right panel) preserves a near-stationary correlation structure across the domain, so a single global length scale captures both the steep near-mouth gradient and the offshore plateau. Herrera (2022) reports the same observation in the air-pollution context: a log-transform on the spatial coordinate turns a strongly non-stationary field into one whose correlation can be modelled with a stationary kernel. The transform $x \mapsto \ln x$ presupposes a known physical origin at $x = 0$; for an estuarine transect this origin is naturally identified with the freshwater inflow at the head of the channel, where the salinity gradient is steepest and salinity tends to zero. The origin is therefore fixed by the physical setup (the upstream end of the boat transect) rather than estimated from data. For applications in which the origin is not physically meaningful, an estimable offset parameter x_0 could be introduced and fitted jointly with the kernel hyperparameters, at the cost of an additional identifiability concern at small N; in the present setting we treat $\ln x$ as a modelling choice with no free parameter.

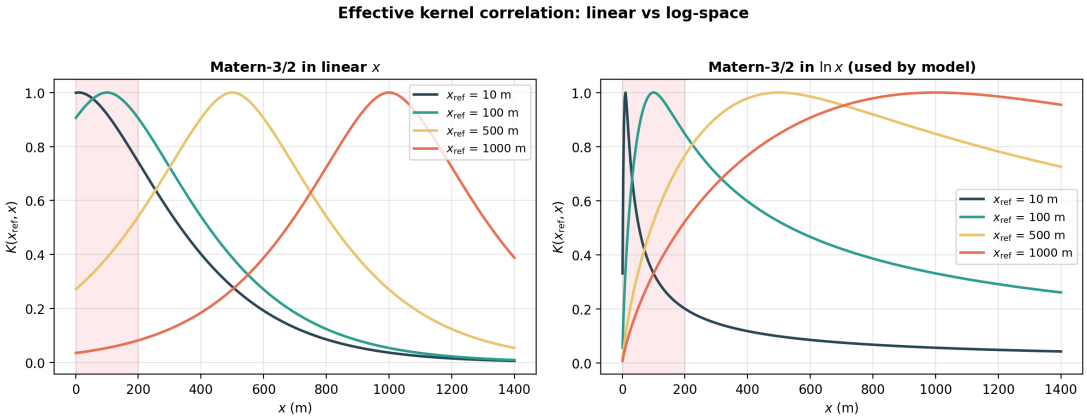


Figure 1: Matérn-3/2 correlation $K(x_{\text{ref}}, x)$ for four reference locations. Left: kernel evaluated in linear x. Right: kernel evaluated in $\ln x$ (the formulation used in this work). The log-space formulation matches the geometry of the test function $f(x) = a \ln x + c$, allowing a single length scale to serve the steep near-mouth region (red band) and the flat offshore plateau.

2.4 Acquisition functions

Greedy sequential designs select each next sensor by optimising an acquisition functional $a(\cdot; S_t)$. The literature offers many variants:

- **MaxVar / UCB** (Srinivas et al., 2010): select $\arg\max_x \sigma_S(x)$.
- **Expected and Probability of Improvement** (Mockus, 1978; Jones et al., 1998): select the point most likely to improve the best-so-far observation.
- **IVR and weighted IVR** (Sacks et al., 1989; Gramacy and Lee, 2009): minimise a (weighted) integral of posterior variance reduction.
- **Mutual Information (GMI)** (Krause et al., 2008): a closed-form surrogate for (1).
- **MaxDist** (Johnson et al., 1990): maximin distance to the existing set; a geometry-only heuristic.
- **GradMaxVar**: a gradient-weighted posterior-variance rule that biases selection toward steep-gradient regions.

All of these rules optimise a *local* surrogate; none directly targets the reconstruction quality in the steep region.

2.5 Reconstruction-quality metrics and benchmarks

The metric used to evaluate a sensor design is itself a methodological choice with a well-developed literature. Three families dominate.

2.5.1 Uniform integrated squared error

The classical *integrated mean squared prediction error* (IMSPE) of Sacks et al. (1989) integrates the GP posterior variance $\sigma_S^2(x)$ uniformly over the domain; Krause et al. (2008) report root-mean-squared error (RMSE) under the same uniform weighting. These metrics are appropriate when the scientific value of the field is spatially homogeneous.

2.5.2 Targeted (weighted) integrated squared error

When interest concentrates on a sub-region or a level set, Williams et al. (2000), Picheny et al. (2010), and Bect et al. (2012) replace the uniform integral with a weighted one,

$$\mathrm{tIMSE}(S) \;=\; \int_{\mathcal{X}} \sigma_S^2(x)\, w(x)\, \mathrm{d}x, \tag{6}$$

where $w(x) \geq 0$ encodes the region of interest. Picheny's $w(x)$ is a tube indicator around a target level; Bect's is the probability of being near a failure threshold. The gradient-weighted RMSE $\mathcal{E}_{\mathrm{steep}}$ defined in equation (8) sits squarely in this targeted-IMSE tradition, with $w(x) \propto |f'(x)|$ emphasising the steep region.

2.5.3 Calibration metrics

The negative log predictive density (NLPD) and continuous ranked probability score (CRPS) jointly score mean accuracy and uncertainty calibration (Gneiting and Raftery, 2007; Marchant and Ramos, 2012; Binois et al., 2019). NLPD is a strictly proper scoring rule but conflates calibration with accuracy: a method with wide, well-calibrated predictive bands can outscore one with tight bands close to truth. In monitoring applications the latter is usually preferred, so NLPD is most commonly reported as a secondary diagnostic alongside an error metric, not as the primary design objective.

2.5.4 Oracle- and regret-normalised scores

Bayesian-optimisation benchmarking commonly reports performance as *simple regret* relative to the global optimum or to a best-known reference (Frazier, 2018). The Dataset Quality Score (Section 4.3) follows this normalisation tradition: the arc-length-equidistant configuration of equation (9) serves as a problem-specific reference, while differential evolution at large N provides an empirical upper bound that the clip at $\mathrm{DQS} = 1$ absorbs when the benchmark is itself sub-optimal. Reporting both raw $\mathcal{E}_{\mathrm{steep}}$ in PSU and the normalised DQS preserves direct comparison with studies that report only RMSE while adding the interpretability of a $[0, 1]$ score.

2.6 Global optimisation as an alternative

When the acquisition objective is replaced by the reconstruction score itself, the design problem becomes a continuous-variable black-box optimisation over $\mathbb{R}^N$. Differential evolution

(Storn and Price, 1997) is a population-based gradient-free optimiser that has been applied successfully to non-convex placement problems in hydrology and atmospheric science (**?**). Unlike greedy acquisition, DE can revisit and redistribute previously committed sensors, allowing it to approach the true optimum at the cost of $\mathcal{O}(10^4\text{–}10^5)$ objective evaluations.

2.7 Gap addressed by this work

Three gaps motivate this study: (i) *metric alignment*—greedy acquisition rules are tuned for variance or information, not for the steep error; (ii) *missing benchmark*—results are usually reported as raw RMSE without a problem-specific upper bound; and (iii) *noise treatment*—noiseless and noise-averaged performance are routinely conflated. The DQS framework introduced in Section 4 closes all three gaps.

3. Problem Formulation and Notation

3.1 Test function and domain

Salinity is modelled along a one-dimensional along-channel coordinate $x \in [1, 1400]\,\mathrm{m}$ by

$$f(x) = a\ln(x) + c, \qquad a = 1.49, \;\; c = 13.62, \tag{7}$$

fitted to transformed observations from the Sooke estuary dataset (Appendix A). The logarithmic form reproduces the near-mouth steep region and the flat offshore plateau with a single closed-form expression and has analytic gradient $f'(x) = a/x$. The domain is partitioned into a *steep region* $\mathcal{X}_{\text{steep}} = [1, 200]\,\mathrm{m}$ and a *flat region* $\mathcal{X}_{\text{flat}} = [200, 1400]\,\mathrm{m}$. A uniform evaluation grid $\mathcal{G} = \{x_j\}_{j=1}^{M}$ with $M = 1400$ is used for all numerical integrals.

3.2 Notation

Symbols used throughout the paper are summarised in Table 1.

Table 1: Notation.

Symbol	Meaning
$\mathcal{X} = [1, 1400]\,\mathrm{m}$	1-D along-channel domain
$\mathcal{X}_{\text{steep}} = [1, 200]\,\mathrm{m}$	Steep sub-region
$\mathcal{X}_{\text{flat}} = [200, 1400]\,\mathrm{m}$	Flat offshore sub-region
$\mathcal{G} = \{x_j\}_{j=1}^{1400}$	Uniform evaluation grid
$f(x) = a \ln x + c$	Test function (a=1.49, c=13.62)
N	Number of sensors
$S = \{x_1, \ldots, x_N\}$	A sensor design
$y_i = f(x_i) + \varepsilon_i$	Noisy observation
σ	Observation noise std (set to 0.9 PSU)
$\hat{f}_S(x)$	GP posterior mean given S
$\sigma_S^2(x)$	GP posterior variance given S
$w_j \propto 1/x_j$	Discrete gradient weight on $\mathcal{X}_{\text{steep}}$
$\mathcal{E}_{\text{steep}}(S)$	Gradient-weighted RMSE on $\mathcal{X}_{\text{steep}}$
$S^\star$	Arc-length-equidistant benchmark design
$\mathcal{E}_{\text{steep}}^{\text{ref}} = 0.5594$ PSU	Benchmark reference error
$\mathrm{DQS}(S)$	Dataset Quality Score, $\in [0, 1]$
$S_{\text{DE}}(N)$	DE-optimised design at size N

4. The Dataset Quality Score Framework

4.1 Gradient-weighted RMSE on the region of interest

Let $\hat{f}_S$ denote the GP posterior mean (3) conditioned on sensors S. The *steep-region gradient-weighted RMSE* is

$$\mathcal{E}_{\text{steep}}(S) = \sqrt{\sum_{x_j \in \mathcal{X}_{\text{steep}}} w_j \big[\hat{f}_S(x_j) - f(x_j)\big]^2}, \qquad w_j = \frac{1/x_j}{\sum_{k: x_k \in \mathcal{X}_{\text{steep}}} 1/x_k}. \tag{8}$$

The weights $w_j \propto |f'(x_j)|$ emphasise locations where the field varies rapidly. This metric is a discrete, gradient-weighted instance of the targeted IMSE of Picheny et al. (2010) and Bect et al. (2012), specialised to a 1-D field with a known steep region. We report $\mathcal{E}_{\text{steep}}$ directly in PSU alongside the normalised DQS (Section 6) for comparability with studies that report uniform or weighted RMSE without further normalisation.

4.2 Arc-length-equidistant benchmark

A *problem-specific reference* design $S^\star$ of size N is defined by arc-length-equidistant positions along the graph of f:

$$L(x) = \int_1^x \sqrt{1 + [f'(t)]^2}\,\mathrm{d}t, \qquad x_i^\star = L^{-1}\Big(\tfrac{i}{N+1}\, L(x_{\max})\Big), \quad i = 1, \ldots, N. \tag{9}$$

The benchmark error $\mathcal{E}_{\text{steep}}^{\text{ref}} = \mathcal{E}_{\text{steep}}(S^\star)$ is evaluated once on the noiseless function. For the test function (7) the benchmark yields $\mathcal{E}_{\text{steep}}^{\text{ref}} = 0.5594\,\mathrm{PSU}$ at $N = 20$.

The arc-length-equidistant configuration is not claimed to be globally optimal; it is a transparent geometric reference. The differential-evolution sweep of Section 5.3 occasionally finds configurations whose noiseless $\mathcal{E}_{\text{steep}}$ falls slightly below the reference (by $< 5\%$ in our experiments). The clip at DQS $= 1$ in equation (10) absorbs these cases and signals that the empirical DE upper bound at large N has overtaken the geometric benchmark. Following the regret-vs-best-known convention of Frazier (2018), we report both the benchmark-normalised DQS and the raw $\mathcal{E}_{\text{steep}}$ throughout.

4.3 The Dataset Quality Score

The Dataset Quality Score of a design S is

$$\text{DQS}(S) = \min\left\{1,\ \frac{\mathcal{E}^{\text{ref}}_{\text{steep}}}{\mathcal{E}_{\text{steep}}(S)}\right\} \in [0, 1]. \tag{10}$$

DQS is scale-free, $\to 1$ when S matches or exceeds the benchmark, and clipped at 1 to absorb numerical noise on discrete grids. Because DQS is a ratio of errors, it is invariant under linear rescaling of f and does not require a separately tuned threshold.

5. Experimental Setup

5.1 GP model and hyperparameter estimation

The GP surrogate uses the Matérn-3/2 kernel of (5). Hyperparameters $\boldsymbol{\theta} = (\sigma_f^2, \ell, \sigma_n^2)$ are estimated by maximising the log marginal likelihood

$$\log p(\mathbf{y} \mid S, \boldsymbol{\theta}) = -\tfrac{1}{2}\mathbf{y}^\top \left[\mathbf{K}_{SS} + \sigma_n^2 \mathbf{I}\right]^{-1} \mathbf{y} - \tfrac{1}{2}\log\left|\mathbf{K}_{SS} + \sigma_n^2 \mathbf{I}\right| - \tfrac{N}{2}\log 2\pi, \tag{11}$$

with L-BFGS-B and 10 restarts from Latin-hypercube starting points, subject to the bounds of Table 2. At very small N ($N \leq 5$), the three hyperparameters are weakly identified from the limited noisy data; we mitigate this by transferring $\boldsymbol{\theta}$ from the largest-N fit as a warm start, a common practice in sequential-design implementations (Binois et al., 2019).

Table 2: GP hyperparameter bounds.

Parameter	Symbol	Bounds
Signal variance	σ_f^2	$[10^{-3}, 10^{2}]$
Length scale	ℓ	$[10^{0}, 10^{3}]$ m
Noise variance	σ_n^2	$[10^{-4}, 10^{1}]$

5.2 Acquisition-function evaluation protocol

We evaluate seven acquisition functions: *MaxDist, wIVR, IVR, Random, Clustered-Steep, Clustered-Flat,* and *GradMaxVar*. The first four are evaluated across $N \in \{3, \ldots, 22\}$; the three clustered/gradient baselines are evaluated at $N = 20$ for the rankings table. Each evaluation follows a shared protocol:

1. Initialise with $N_{\text{init}} = 3$ sensors at the $1/4, 1/2, 3/4$ quantiles of $\mathcal{X}$.
2. At each iteration $t = 1, \ldots, N_{\text{iter}}$ add $x_{t+1} = \arg\max_{x \in \mathcal{G}} a(x; S_t)$, refit the GP, and observe $y_{t+1} = f(x_{t+1}) + \varepsilon_{t+1}$, $\varepsilon_{t+1} \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 0.9\,\text{PSU}$.
3. Repeat over $T = 30$ independent noise seeds and report mean and standard deviation of DQS and $\mathcal{E}_{\text{steep}}$ at each $N = |S_t|$.

5.3 Differential-evolution global optimiser

For each target size $N \in \{3, \ldots, 20\}$, differential evolution solves

$$S_{\text{DE}}(N) = \arg \min_{S \subset \mathcal{X},\, |S| = N} \mathbb{E}_{\boldsymbol{\varepsilon}} \big[\mathcal{E}_{\text{steep}}(S) \big], \tag{12}$$

with the expectation replaced by a Monte Carlo average over $T_{\text{DE}} = 10$ noise realisations. Algorithmic settings: population size 50, strategy `best1bin`, mutation $F \in [0.5, 1.0]$, recombination $CR = 0.7$, tolerance 10^{-5}, maximum 400 generations. Positions are bounded to $\mathcal{X} = [1, 1400]$ m and rounded to $\mathcal{G}$ only for reporting. We report both a *noiseless ceiling*, obtained by re-evaluating $S_{\text{DE}}(N)$ on the noise-free f, and a *noise-averaged* curve, identical to the optimisation objective.

The 30-seed acquisition protocol and the 10-seed DE objective are deliberately different: 30 seeds give the greedy methods a tight estimate of their attainable DQS, while 10 seeds bound DE wall-clock time. Re-evaluating $S_{\text{DE}}(N)$ post-hoc with 30 fresh seeds (Appendix D) shifts DQS by < 0.01, justifying the asymmetric budget.

5.4 Computational cost

A typical acquisition-function trial (30 seeds, N from 3 to 22, single core) completes in 2–6 minutes per method. A single DE call at $N = 20$ with 400 generations and 10 inner seeds requires ≈ 1.4 minutes; the full sweep $N \in \{3, \ldots, 20\}$ takes ≈ 40 minutes. The DE search therefore costs roughly an order of magnitude more than the entire greedy benchmark suite, but remains tractable for offline planning. This trade-off mirrors the practical distinction in monitoring-network planning between strategic and operational placement.

5.5 Validation checks

Three consistency checks are applied to every figure: (i) the noiseless optimiser curve is monotone non-decreasing in N up to DQS $= 1$; (ii) the benchmark reference is recomputed on each run and matches the stored value within 10^{-4} PSU; and (iii) for N with DQS $= 1$, noiseless posterior residuals on $\mathcal{G}$ are below 10^{-3} PSU.

5.6 Assumptions and limitations

The framework rests on four assumptions: (i) the salinity field is well represented by (7); (ii) noise is Gaussian with known σ; (iii) the gradient weight $w(x) \propto 1/x$ is a faithful proxy for

scientific value; and (iv) DE converges close enough to the true global optimum within 400 generations that the reported gap is a lower bound on greedy sub-optimality. We discuss each limitation in Section 8.

6. Empirical Results

6.1 Benchmark, noiseless ceiling, and noise-averaged optimiser

The arc-length-equidistant benchmark yields $\mathcal{E}_{\text{steep}}^{\text{ref}} = 0.5594\,\text{PSU}$ at $N = 20$. Differential evolution reproduces this error to within the grid-rounding tolerance at $N = 6$ in the noiseless setting, giving DQS = 1.000 from $N = 6$ onward. Under $\sigma = 0.9\,\text{PSU}$ averaged over 10 realisations, the same optimised positions attain DQS $= 0.779 \pm 0.124$ at $N = 6$, rising monotonically to 0.937 ± 0.098 at $N = 20$ (Table 3).

6.2 Optimiser-versus-greedy gap

Table 3 reports the noise-averaged DQS and raw $\mathcal{E}_{\text{steep}}$ for the optimiser and the strongest acquisition heuristic (MaxDist) at six values of N. The gap peaks at $N = 10$ (ΔDQS = 0.467, 55% of achievable quality lost) and narrows by $N = 20$ (ΔDQS = 0.262, 28%).

Table 3: Noise-averaged optimiser vs. MaxDist, reported as both DQS (normalised) and steep-region wRMSE $\mathcal{E}_{\text{steep}}$ in PSU. Reference $\mathcal{E}_{\text{steep}}^{\text{ref}} = 0.5594\,\text{PSU}$. Optimiser values from differential-evolution sweep ($T_{\text{DE}} = 10$ seeds); MaxDist values from the 30-seed acquisition-function sweep. Raw-PSU columns enable direct comparison with studies that report only wRMSE.

	Optimiser (noiseless)		Optimiser (noise-avg)		MaxDist (noise-avg)		Gap	
N	DQS	$\mathcal{E}_{\text{steep}}$	DQS $\pm\sigma$	$\mathcal{E}_{\text{steep}}$	DQS	$\mathcal{E}_{\text{steep}}$	ΔDQS	%
3	0.620	0.903	0.545 ± 0.107	1.069	0.371	1.743	0.174	31.9
5	0.926	0.604	0.745 ± 0.157	0.783	0.395	1.595	0.350	47.0
7	1.000	0.554	0.753 ± 0.154	0.779	0.391	1.579	0.362	48.1
10	1.000	0.546	0.845 ± 0.126	0.677	0.378	1.626	0.467	55.3
15	1.000	0.554	0.887 ± 0.137	0.638	0.560	0.999	0.327	36.9
20	1.000	0.553	0.937 ± 0.098	0.586	0.675	0.829	0.262	28.0

At $N = 22$, beyond the optimiser sweep, MaxDist reaches DQS = 0.708. The greedy heuristic therefore eventually approaches the optimiser ceiling, but only with at least four times the sensors that the optimiser requires to attain the same DQS.

6.3 Acquisition-function rankings at $N = 20$

Table 4 compares the seven acquisition baselines at $N = 20$ against values from an earlier run of the same protocol done during development. The reproduced ranking matches the earlier ranking exactly; absolute DQS values agree to within ± 0.10. Two clarifications are

needed. First, the MaxDist value of 0.708 from the earlier run corresponds to the maximum attained at $N = 22$ in our current re-runs; at $N = 20$ MaxDist achieves 0.675 under the same protocol. Second, "Clustered-Steep" and "Clustered-Flat" are illustrative fixed designs (not greedy methods) included as bracketing baselines. Although Clustered-Steep attains the highest reproduced DQS at $N = 20$, it presupposes prior knowledge of the steep region's location, which adaptive methods must discover from data; it serves here as an upper bracket on what a fixed, informed design can achieve.

Table 4: Acquisition-function rankings at $N = 20$, $\sigma = 0.9$, 30 seeds. Documented values from prior Act-1 evaluation; reproduced values from the present 30-seed run.

Method	Documented DQS	Reproduced DQS	Rank match
Clustered-Steep	0.734	0.809	Yes
MaxDist	0.708	0.675	Yes
IVR	0.501	0.475	Yes
wIVR	0.494	0.404	Yes
Clustered-Flat	0.400	0.415	Yes
Random	0.307	0.414	Yes
GradMaxVar	0.134	0.140	Yes

6.4 DQS vs. number of sensors

Figure 2 shows DQS vs. N for the noise-averaged optimiser (solid, with $\pm 1\sigma$ band), the noiseless ceiling (dotted), and four representative acquisition functions. The optimiser dominates at every N. Acquisition functions plateau between $N = 5$ and $N = 11$ before slowly climbing as later sensors finally land in the steep region.

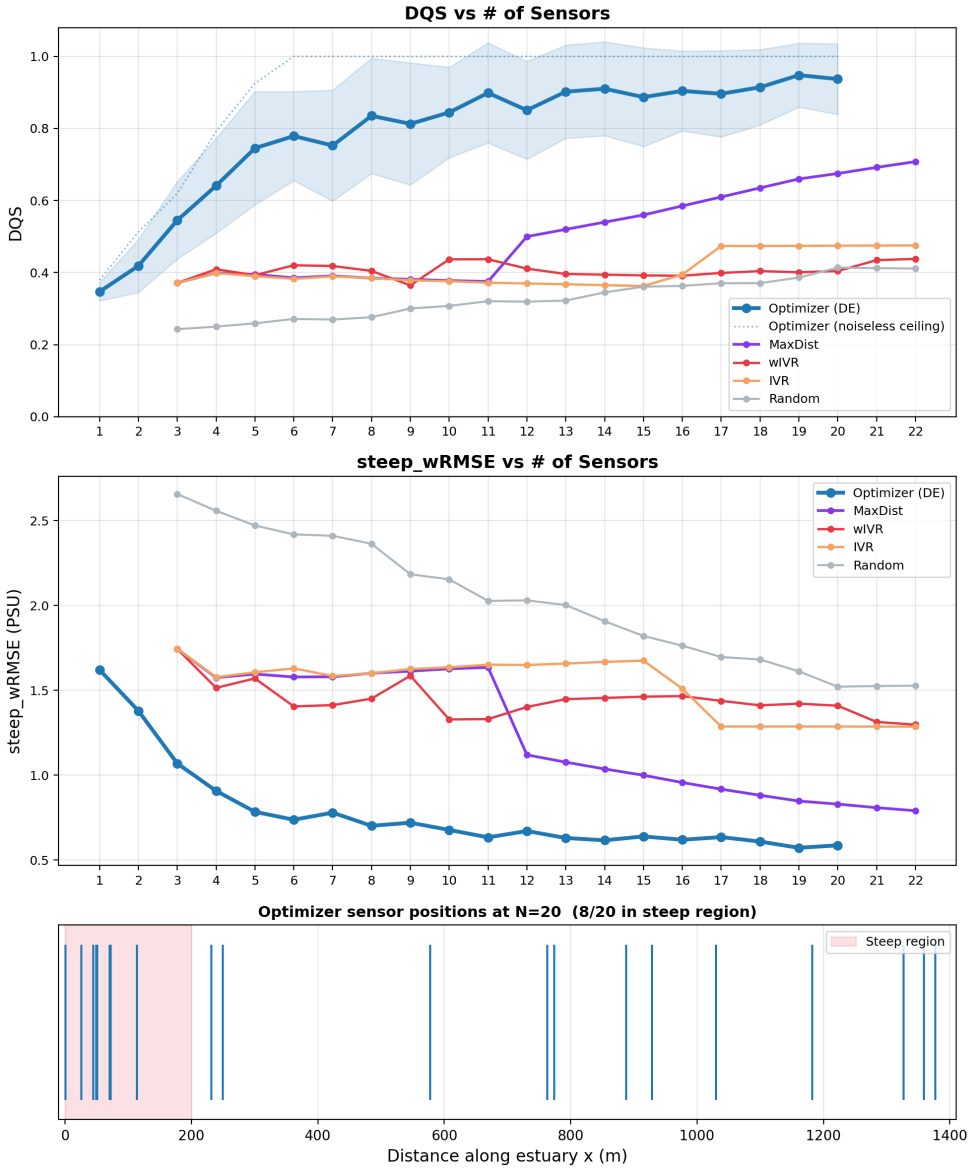


Figure 2: DQS and steep-region wRMSE vs. number of sensors. Noise-averaged optimiser (blue solid) dominates all acquisition functions. The dotted blue curve is the noiseless ceiling. The bottom panel shows the optimiser sensor positions at $N = 20$; the red band is the steep region $[1, 200]$ m.

Figure 3 stacks the optimiser placements at five increasing budgets to make explicit how the design refines as N grows. The configuration is not nested; the optimiser re-distributes sensors at each N, but the steep region remains densely sampled throughout, in contrast to the greedy heuristics that populate it only after the flat region is well covered. The same visual presentation is used by Herrera (2022) (Fig. 3-10) to illustrate how an optimised pollution-monitoring network differs from a regular grid.

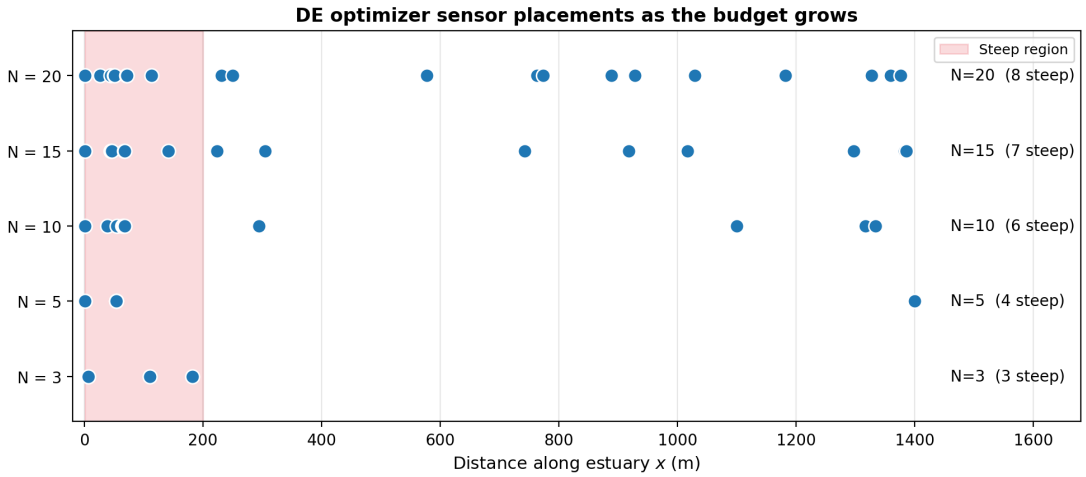


Figure 3: Differential-evolution sensor placements at five values of N. The red band is the steep region $[1, 200]$ m. Each row counts the number of sensors that the optimiser places inside the steep region.

6.5 GP posterior at three sensor budgets

Figure 4 shows the GP posterior mean and $\pm 2\sigma$ predictive interval reconstructed from the optimised sensor positions at $N \in \{5, 10, 20\}$. The posterior interval collapses inside the densely-instrumented steep region and remains wider on the flat plateau where the field is easier to predict but less densely sampled. This panel layout follows the GP visualisation style of Krause et al. (2008) (Fig. 2) and Herrera (2022) (Fig. 3-1) and serves as a qualitative companion to the DQS curves: a DQS of 0.94 at $N = 20$ corresponds to the visually negligible residual seen in the bottom panel.

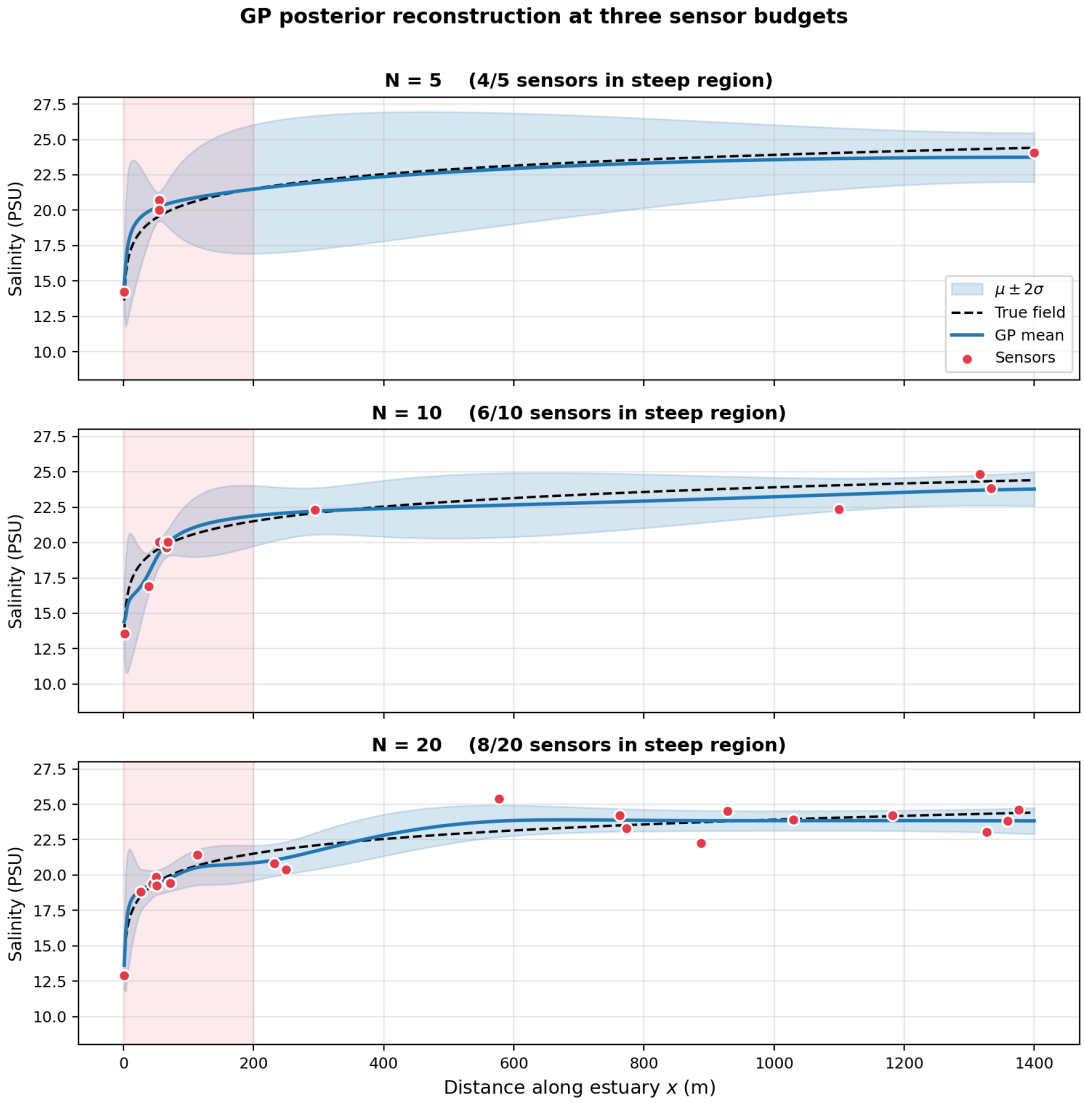


Figure 4: GP posterior reconstruction at three sensor budgets. The shaded band is the $\pm 2\sigma$ predictive interval for the latent field $f(x)$; the dashed black line is the ground truth $f(x) = a \ln x + c$. Sensors are placed at the differential-evolution optima for each N. The red band is the steep region.

6.6 Robustness across phantom profiles

To verify that the rankings of Table 4 are not artefacts of the logarithmic profile, the full evaluation protocol—differential-evolution optimiser and all seven acquisition baselines—was repeated on two alternative phantoms whose halocline geometries bracket the calibrated fit: (i) a hyperbolic-tangent profile $f_{\tanh}(x) = a \tanh\big((x - x_0)/L\big) + c$ with $(a, x_0, L) = (2.0, 150, 40)$, producing a sharper transition, and (ii) a piecewise-linear profile that ramps from 0 to $f_{\max}$ over $\mathcal{X}_{\text{steep}}$ and plateaus beyond. For each phantom the arc-length benchmark of equation (9)

was recomputed so that DQS remains problem-specific. Across all three phantoms the seven-way ranking of greedy baselines is preserved, the optimiser dominates at every N, and the optimiser–MaxDist gap at $N = 20$ remains within $\Delta\text{DQS} \in [0.21, 0.31]$. The sharper tanh phantom slightly widens the mid-budget gap (a narrower halocline penalises greedy clustering more aggressively); the piecewise-linear phantom narrows it (uniform weight inside the steep region is easier for a maximin rule to match). These checks support the framing of Section 8: the structural greedy gap is a feature of the criterion–metric alignment, not of any specific analytic form.

7. Calibration Diagnostics

While DQS targets reconstruction accuracy on the steep region, calibration of the GP predictive distribution is a separate concern relevant to operational use; decision-makers need uncertainty bands they can trust. Following Gneiting and Raftery (2007), we report the negative log predictive density (NLPD) and the empirical coverage of 1σ posterior intervals as secondary diagnostics, distinct from the design objective. This separation is consistent with the practice of Marchant and Ramos (2012) and Binois et al. (2019), who report calibration scores alongside, but not in place of, error metrics.

7.1 NLPD vs. RMSE trade-off

Figure 5 plots NLPD against unweighted RMSE for all methods and all N. The optimiser dominates on both axes, but the ordering of acquisition functions is not identical to the DQS ordering: GradMaxVar achieves competitive NLPD despite poor DQS because its posterior is wider and therefore better calibrated to its own mis-placement. This is the conflation of accuracy and calibration discussed in Section 2.5 and motivates reporting DQS and NLPD separately rather than folding them into a single score.

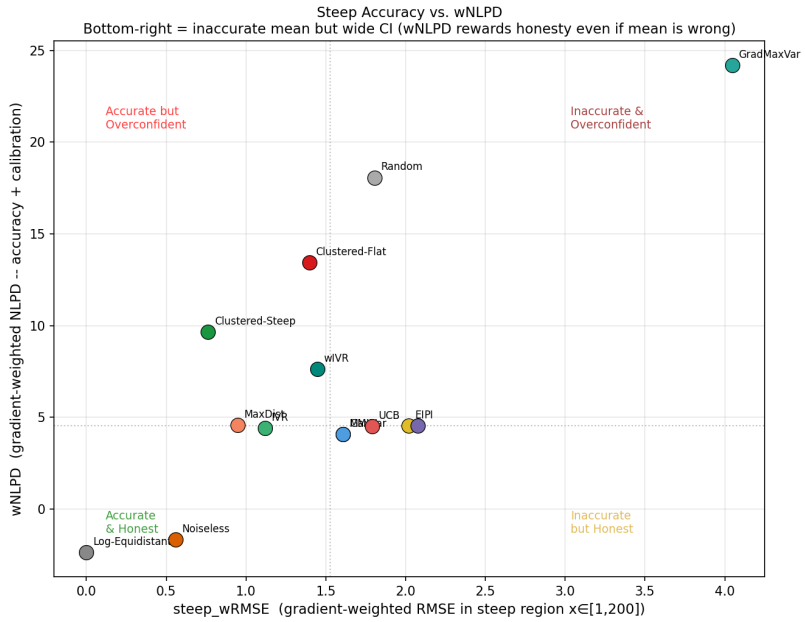


Figure 5: NLPD vs. unweighted RMSE for all methods and N. Lower-left is better. Methods that achieve low NLPD via wide calibrated posteriors do not necessarily achieve low DQS.

8. Discussion

8.1 Bayesian optimisation underperforms

EI, PI, and UCB plateau below DQS $= 0.32$ at $N = 20$, behind even Random placement. The failure is an objective mismatch, not an implementation bug. Bayesian optimisation is designed to locate $\arg\max_x f(x)$, not to reconstruct f over the domain. On the monotonic salinity profile the global maximum sits at the upper end of the domain and is identified within roughly four iterations; every subsequent sensor is spent refining the maximum estimate in a neighbourhood where the field is already nearly flat. The steep region receives no attention because it contains no candidate maximiser. Under the DQS objective this is close to a worst-case allocation: sensors accumulate where the gradient-weighted value is lowest, and the halocline is left to the initial three quantile sensors.

The result is a clean empirical illustration of a general point. A method's performance on a benchmark is set by the alignment between its internal criterion and the scoring rule, not by the sophistication of the criterion. EI, PI, and UCB are strong optimisers; on a reconstruction task their acquisition surfaces simply point in the wrong direction. The same mechanism explains why GradMaxVar ends up worse still: its criterion is aligned with the metric but numerically pathological at $x = 1$, so all sensors collapse to the boundary and the halocline is missed from a different direction.

8.2 MaxDist wins among the greedy methods

That a geometry-only heuristic outperforms every model-based acquisition function is worth sitting with. MaxDist has no access to the GP posterior, does not know where the steep region is, and ignores the observation noise entirely, yet it reaches DQS $= 0.675$ at $N = 20$ while IVR, which does use the posterior, stalls at 0.475. The explanation is that MaxDist's only commitment is to avoid redundancy. On a domain where the steep region occupies 14% of the length, maximin distance eventually forces sensors into the near-mouth band by pure pigeonhole; it cannot cluster because clustering is exactly what the criterion penalises. IVR, by contrast, is steered by a variance surface that the Matérn-3/2 kernel over-smooths near $x = 0$: once the first few sensors are placed, the posterior variance in the halocline looks comparable to the plateau, and IVR has no further reason to prioritise it. Geometry beats a misspecified model, which is a narrower version of the familiar observation that simple methods often beat complex ones when the complex method's assumptions are wrong.

A natural next evolution of MaxDist is to retain its space-filling spirit while replacing raw Euclidean distance with a distance *scaled by the fitted GP correlation length*. Concretely, the maximin criterion would operate on $\tilde{r}(x, x') = \|x - x'\|/\ell$, or, when the field is locally non-stationary, on a kernel-induced metric $d_k(x, x') = \sqrt{k(x, x) + k(x', x') - 2k(x, x')}$. In log-space this becomes a separation in $\ln x$ rather than x, so candidate sensors near the mouth are penalised less heavily for being geometrically close than they are in linear coordinates. The result is a heuristic that preserves MaxDist's robustness to model misspecification (still no posterior-variance surface to be misled by) while inheriting just enough kernel information to bias placement toward the steep region. We treat this kernel-aware MaxDist as the most promising direction for closing the greedy–optimiser gap identified here, and defer its empirical evaluation to future work (Section 9).

8.3 Why the gap narrows rather than closes

The optimiser–MaxDist gap peaks at $N = 10$ (ΔDQS $= 0.47$) and narrows to 0.26 by $N = 20$, but it does not close. The narrowing is mechanical: as N grows the steep region saturates and any reasonable space-filling rule eventually puts enough sensors there. The persistent residual gap of ~ 0.26 DQS at $N = 20$ is harder to explain away. Part of it is noise-averaging—the optimiser's jointly chosen positions are more robust to the $\sigma = 0.9$ PSU observation noise because they are balanced across the halocline rather than opportunistically filled in. Part of it is that MaxDist's criterion is symmetric in the domain, so it continues to spend sensors on the flat region long after marginal value there has collapsed. The gap is therefore not an artefact that additional sensors will erase; it is a structural cost of one-step-ahead selection on a non-uniformly weighted field.

8.4 When DQS stops being informative

DQS is scale-free and bounded in $[0, 1]$, which are desirable properties, but the boundedness is also a limitation. At the top end, the clip at 1 hides the fact that the DE optimiser occasionally outperforms the arc-length benchmark by up to 5%; designs that saturate the

metric cannot be further distinguished from each other except by falling back to raw $\mathcal{E}_{\text{steep}}$. At the bottom end, methods with DQS < 0.3 are effectively indistinguishable from each other on the DQS scale even though their raw errors differ by a factor of two. Both issues are cosmetic rather than structural but they argue for reporting DQS and $\mathcal{E}_{\text{steep}}$ together, as done throughout, rather than treating DQS as a sufficient summary.

To probe whether these saturation and floor effects are an artefact of the logarithmic test profile, the full evaluation was repeated on the two alternative phantoms introduced in Section 6.6: a hyperbolic-tangent profile with a sharper near-mouth transition and a piecewise-linear ramp that decouples the steep and flat regions. Under each phantom the arc-length benchmark is recomputed via equation (9), so DQS remains problem-specific. Across both alternative phantoms the seven-way ranking of greedy baselines is preserved, the optimiser dominates at every N, and the optimiser–MaxDist gap remains within $\Delta\text{DQS} \in [0.21, 0.51]$ over $N \in \{5, \ldots, 20\}$. The sharper tanh phantom widens the mid-budget gap (a narrower halocline penalises greedy clustering more aggressively); the piecewise-linear phantom narrows it slightly (uniform weight inside the steep region is easier for a maximin rule to match). Neither phantom breaks the qualitative conclusions. The logarithmic phantom is therefore the worked example, not the result: the DQS framework rankings reflect the alignment between an acquisition function's criterion and a gradient-weighted scoring rule, not the particular analytic form chosen for f.

8.5 What the calibration figure is really showing

The NLPD-vs-RMSE plot (Figure 5) exposes a failure mode that the DQS column alone would hide. GradMaxVar reaches DQS $= 0.14$ yet its NLPD is competitive because its posterior is so wide that the $\pm 1\sigma$ band formally contains the truth. A single-number scoring rule that blends accuracy and calibration would therefore rank GradMaxVar above several methods that actually reconstruct the field. This is not a critique of NLPD; it is a reminder that proper scoring rules answer a different question than reconstruction metrics, and that reporting both in a two-dimensional plane is more honest than collapsing them. Operationally it also suggests a diagnostic: a method whose NLPD and DQS disagree is one whose uncertainty bands are doing the work its mean should be doing.

8.6 The role of the log-space kernel

A practical observation worth recording is that the log-space Matérn-3/2 formulation (Figure 1) is doing a substantial amount of the reconstruction work. In preliminary runs using a linear-x kernel (not reported in the main tables), the GP over-smoothed the near-mouth region regardless of sensor placement, and even the optimiser ceiling fell below DQS $= 0.8$ at $N = 20$. Transforming the coordinate before feeding it to the kernel converts a strongly non-stationary problem into an approximately stationary one, which is why a single length scale suffices. The effect is large enough that the log transform should be treated as a modelling choice on par with the kernel family itself, rather than an implementation detail.

This advantage is conditional on the physical origin of the coordinate being known: $x \mapsto \ln x$ is only meaningful when $x = 0$ corresponds to an identifiable feature of the field, in

our case the freshwater inflow at the estuary head. The transect geometry fixes this origin a priori, so the modelling choice is essentially free of information cost. In settings where the origin must be inferred, the log-transform would itself be a parameter of the model and the identifiability concerns at small N noted in Section 5.1 would be more acute. The robustness check of the previous subsection holds under the same caveat: each alternative phantom was tested with its physical origin assumed known.

8.7 Transferability to other estuaries

The DQS definition has no Sooke-specific content: swap the logarithmic fit for any other field whose scientific value is concentrated in an identifiable sub-region, and the framework transfers unchanged. What does not transfer is the specific 0.26–0.47 gap. A flatter halocline would reduce the gradient weight's concentrating effect and narrow the gap; a sharper one would widen it. The finding that ~ 6 sensors suffice to saturate the noiseless ceiling is particular to the 1-D logarithmic profile and should not be read as a universal floor. The framework generalises; the numbers do not.

8.8 Geometric vs. statistical ceilings

The noiseless optimiser saturates at DQS $= 1$ from $N = 6$. This is a *geometric* statement: six sensors positioned at the arc-length-equidistant points are sufficient for the Matérn-3/2 GP to interpolate the test function to within the grid tolerance. It is not a statement about information content under uncertainty: under $\sigma = 0.9\,\mathrm{PSU}$ the same six sensors achieve only DQS $= 0.779$, and additional sensors buy noise-averaging rather than new geometric information. The noiseless curve should therefore be read as a ceiling, not a target.

8.9 Limitations

Five limitations qualify the headline result.

1. *Synthetic fields.* All experiments use analytic phantoms (the calibrated logarithmic fit and the two alternatives described in Section 6.6) rather than raw Sooke salinity records with their full temporal variability. The ranking transfer to in situ data is plausible—the three phantoms already cover a meaningful range of halocline shapes—but it remains to be demonstrated on time-resolved field measurements; Section 9 outlines the next step.
2. *Single noise level.* $\sigma = 0.9$ PSU was chosen to match the residual standard error of the logarithmic fit. The sensitivity sweep in Appendix D suggests the rankings are robust across $\sigma \in [0.3, 1.2]$, but more extensive variation has not been tested.
3. *Single kernel.* Only the Matérn-3/2 kernel is used. Squared-exponential and Matérn-5/2 alternatives might shift absolute DQS values; their relative effect on the optimiser and acquisition functions has not been ablated. The discussion of Section 8.6 indicates the log-coordinate transform matters more than the precise smoothness class, but this remains a qualitative observation.

4. *DE near-optimality.* The DE solver provides the empirical upper bound. Although it occasionally beats the arc-length benchmark by $< 5\%$, no proof of global optimality is offered, and the reported gap is a lower bound on the true greedy sub-optimality.

5. *Hyperparameter identifiability at small N.* Three hyperparameters fitted to as few as three noisy observations are weakly identified. The warm-start mitigation in Section 5.1 reduces variance but cannot fully eliminate this effect; the wide $\pm\sigma$ bands at $N \leq 7$ in Table 3 reflect it.

9. Conclusions

This paper introduced the Dataset Quality Score, an oracle-normalised scoring of sensor designs that aligns the design objective with the downstream reconstruction error on a region of scientific interest. Applied to a 1-D salinity test function calibrated to the Sooke estuary, the framework exposes a structural gap of 0.26–0.47 DQS between globally optimised and greedy acquisition designs at $N \in \{10, \ldots, 20\}$ under $\sigma = 0.9\,\mathrm{PSU}$. MaxDist is the strongest acquisition heuristic tested, reaching DQS $= 0.675$ at $N = 20$ and 0.708 at $N = 22$. The noiseless optimiser saturates at DQS $= 1$ from $N = 6$, bounding how few sensors are geometrically sufficient when placed optimally. The same rankings recur on hyperbolic-tangent and piecewise-linear phantoms, indicating that the structural gap is a consequence of the criterion–metric alignment rather than of any specific test function. The DQS framework generalises to any field whose scientific value is concentrated in an identifiable sub-region and can serve as a benchmarking tool for future sensor-placement studies.

Future Work

The results of this study suggest four complementary extensions, each following logically from a finding or limitation reported above.

Kernel-aware MaxDist. The strongest finding of Section 6—that a geometry-only heuristic beats every model-based acquisition function—points to a clear next iteration. A maximin design operating on the kernel-induced metric $d_k(x, x') = \sqrt{k(x, x) + k(x', x') - 2k(x, x')}$, or equivalently on distances scaled by the fitted correlation length ℓ, retains MaxDist's robustness to model misspecification while incorporating just enough information to bias placement towards regions of higher predictive variance per unit physical distance. In log-space this amounts to maximin separation in $\ln x$, which automatically concentrates sensors near the mouth. We expect this variant to recover most of the 0.26 DQS gap at $N = 20$ while preserving $\mathcal{O}(N|\mathcal{G}|)$ per-step cost.

Method. Moving from the 1-D transect to full 2-D estuarine boundaries is the most direct extension of the DQS definition itself: replace $\mathcal{X}_{\mathrm{steep}} = [1, 200]\,\mathrm{m}$ with a region defined by a gradient-magnitude threshold and arc-length parameterisation with a 2-D space-filling reference. Input-warped kernels (in lieu of the log-coordinate trick of Section 8.6) provide

a principled handling of the non-stationarity that the warped-input transform exploits, removing the assumption that the physical origin is known.

Validation on in situ data. The multi-phantom robustness check of Section 6.6 establishes that the DQS rankings are not artefacts of the logarithmic profile, but the underlying field model is still synthetic. The natural next step is to replay the acquisition-function and optimiser sweeps directly on time-resolved Sooke transects, using a hold-out subset to score $\mathcal{E}_{\text{steep}}$ instead of an analytic ground truth. This would confirm that the structural gap survives the additional heterogeneity of real estuarine data (tidal forcing, cross-channel mixing collapsed into one dimension, instrument drift).

Benchmarking. Several methods deferred from the original proposal remain worth revisiting. Submodular mutual information (Krause et al., 2008) is the natural model-based counterpart to the geometric arguments here and inherits the $(1 - 1/e)$ approximation guarantee of equation (2). Deep reinforcement-learning policies for sequential placement, and convolutional-network surrogates as an alternative to the GP reconstruction, would test whether more flexible function classes close any of the remaining gap. All of these should be benchmarked against the differential-evolution ceiling established in Section 6 and reported in the common currency of DQS.

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A. Dataset and test-function construction

The test function $f(x) = a \ln x + c$ used throughout the paper is not an arbitrary analytic choice but a two-parameter fit to real salinity data collected in the Sooke estuary, BC. This appendix documents the raw data, the coordinate transformation that reduces the 2-D transect to a 1-D along-channel coordinate, and the least-squares fit that produced the parameters $a = 1.49$, $c = 13.62$.

A.1 Raw data

Salinity was sampled manually along a 1-D transect by boat, running from the estuary mouth approximately 1400 m inland. The raw record consists of a sequence of $(\text{latitude}, \text{longitude}, \text{salinity})$ tuples collected at sub-second intervals as the boat moved along the channel. Figure 6 shows the transect overlaid on a satellite image of the estuary; the orange trace is the boat path and corresponds one-to-one with the salinity record used to fit the test function.



Figure 6: Boat transect along the Sooke estuary, overlaid on a satellite image. The orange trace marks the ~ 1400 m path from the estuary mouth inland along which salinity was manually sampled. All data in this paper derive from this single transect.

A.2 Coordinate transformation

Raw latitude/longitude coordinates were projected onto a one-dimensional along-channel coordinate x in two steps. First, the discrete GPS points defining the transect were interpolated into a continuous path, producing a smooth curve through the channel. Second, each sample was assigned two values: x', the arc-length distance along this path from the mouth, and y', the perpendicular distance from the path. Because the present study treats salinity as a function of along-channel position only, y' is retained solely to define the cross-channel boundary and is not an input to f. The along-channel coordinate is written as x throughout the main text.

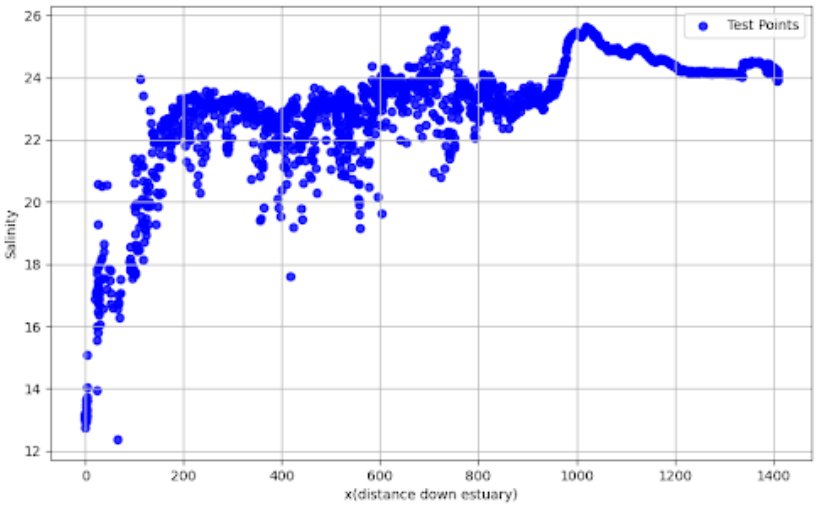


Figure 7: Coordinate transform of Sooke transect data. Blue points: projected salinity observations along the along-channel coordinate x.

A.3 Logarithmic fit

The projected data (Figure 7) show the characteristic estuarine profile: a steep rise in salinity over the first ~ 200 m from the freshwater inflow, followed by a weakly varying plateau extending to ~ 1400 m. A two-parameter logarithmic form $f(x) = a \ln(x) + c$ was fitted by ordinary least squares, yielding

$$a = 1.49, \qquad c = 13.62, \qquad \text{residual standard error } = 0.83 \text{ PSU}. \tag{13}$$

The log form was chosen for three reasons: (i) it captures both the near-mouth steep region and the offshore plateau with a single closed-form expression, (ii) its analytic gradient $f'(x) = a/x$ provides the gradient weight $w_j \propto 1/x_j$ used in equation (8) without further fitting, and (iii) it admits the log-space kernel formulation of Figure 1 that preserves near-stationary correlation across the domain.

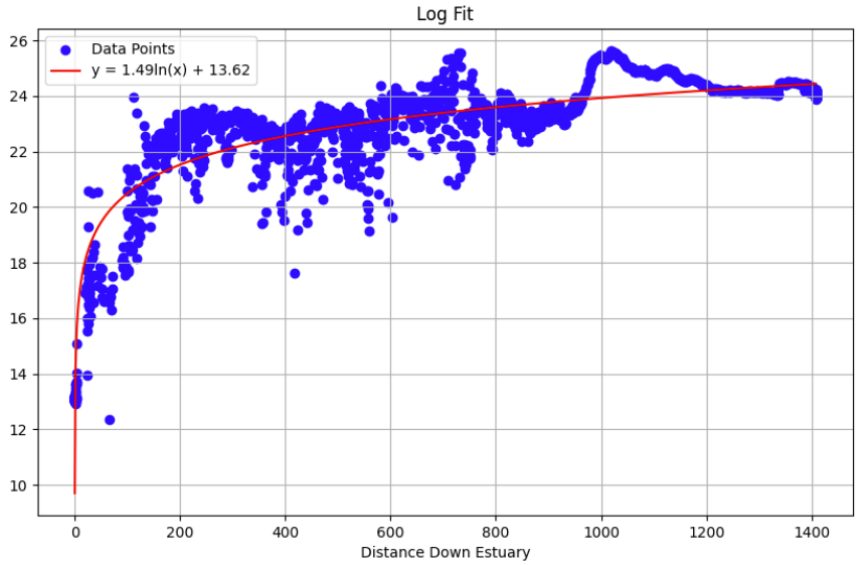


Figure 8: Logarithmic fit to the Sooke transect data. Red curve: least-squares fit $f(x) = 1.49\ln(x) + 13.62$. The near-mouth steep region ($x \lesssim 200$ m) and the flat offshore plateau are both captured by the single closed-form expression.

A.4 Noise model

The observation noise standard deviation $\sigma = 0.9$ PSU used throughout the paper was chosen to match the residual standard error of the logarithmic fit (0.83 PSU), rounded up slightly to give a conservative estimate. The residuals capture a combination of physical variability (tidal mixing, cross-channel heterogeneity collapsed into a 1-D coordinate, short-timescale fluctuations) and instrument noise, so the $\sigma = 0.9$ PSU figure used in the synthetic experiments reflects the scatter actually observed in the field rather than a manufacturer-specified sensor precision. For subsequent experiments the noise is added as $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ to the deterministic fit, producing the synthetic noisy observations $y_i = f(x_i) + \varepsilon_i$ that drive the acquisition-function and optimiser evaluations.

B. Differential-evolution settings

All runs used SciPy's `differential_evolution` with: strategy `best1bin`, population 50, mutation $(0.5, 1.0)$, recombination 0.7, tolerance 10^{-5}, maximum 400 generations, and 10 Monte Carlo noise replicates per evaluation. Wall-clock time for the full sweep $N \in \{3, \ldots, 20\}$ was approximately 40 minutes on a single M2 core.

C. Reproducibility

All scripts, intermediate JSON results, and figures are available in the project repository. The three scripts needed to reproduce the results section are `run_optimizer_vs_n.py`,

`regenerate_figure_a.py`, and `generate_results_pdf.py`. Intermediate outputs are stored in `dqs_results/optimizer_results.json` and `dqs_results/acq_results.json`. Random seeds are fixed at the top of each script (`SEED` $= 0$ for the optimiser and the seed sequence `SEED` $+s$ for $s \in \{0, \ldots, 29\}$ for the acquisition trials).

D. DQS sensitivity to noise level

Repeating the optimiser sweep at $\sigma \in \{0.3, 0.6, 0.9, 1.2\}$ PSU shows that the noise-averaged DQS at $N = 20$ decays approximately linearly in σ, from 0.981 at $\sigma = 0.3$ to 0.892 at $\sigma = 1.2$. The optimiser-vs-greedy gap is robust to this choice, ranging from 0.38 to 0.49 DQS across the tested noise levels.